Technology Stack Template: India's Agricultural Crop Production Analysis

Date: 30 June 2026

Team ID: XXXX

Project Name: India's Agricultural Crop Production Analysis

Maximum Marks: 4 Marks

Technical Architecture Diagram

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|   Data Sources       |      |    Processing Layer   |      |   Analytics & Dashboard|
| (Data.gov.in, FAO)   |------>| (Spark / Python / ML) |------>| (Streamlit / PowerBI)  |
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|  External APIs       |      |    Database Storage   |      |    Model Deployment   |
| (Weather, Market)    |      | (PostgreSQL / AWS RDS) |      | (Docker / Kubernetes)  |
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Technology Stack

S.No	Component	Description	Technology
1	User Interface	Dashboard for crop yield analysis	React.js, Tailwind CSS
2	Application Logic-1	Data extraction from Govt portals	Python (BeautifulSoup, Requests)
3	Application Logic-2	Crop yield predictive analytics	Scikit-Learn (Random Forest)
4	Application Logic-3	Time-series crop forecasting	Prophet/ARIMA
5	Database	Local production database	PostgreSQL
6	Cloud Database	Scalable cloud storage	AWS RDS
7	File Storage	Raw agricultural datasets	AWS S3

8	External API-1	Weather-Crop correlation	OpenWeatherMap API
9	External API-2	Market price integration	Agmarknet API
10	Machine Learning Model	Yield Prediction engine	TensorFlow/Scikit-Learn
11	Infrastructure	Cloud deployment platform	Docker, Kubernetes

Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Data manipulation & Analysis	Python (Pandas, NumPy)
2	Security Implementations	User access and data privacy	JWT, OAuth 2.0
3	Scalable Architecture	Cloud-native design	Microservices on AWS
4	Availability	Load balancing	AWS Elastic Load Balancer
5	Performance	Efficient data query caching	Redis

References

- https://data.gov.in/ (Agricultural datasets for India)
- https://agricoop.nic.in/ (Ministry of Agriculture & Farmers Welfare)
- https://www.fao.org/faostat/en/ (Global Agricultural Statistical Data)
- https://www.analyticsvidhya.com/blog/2021/07/crop-yield-prediction-using-machine-learning/